从零开始搭建一个项目

Huanghai

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COGNEX

培训内容

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4. 标定通讯协议 - 对位标定
5. Framework 设计
6. Inspection 设计
7. 生产通讯协议
8. Inspection

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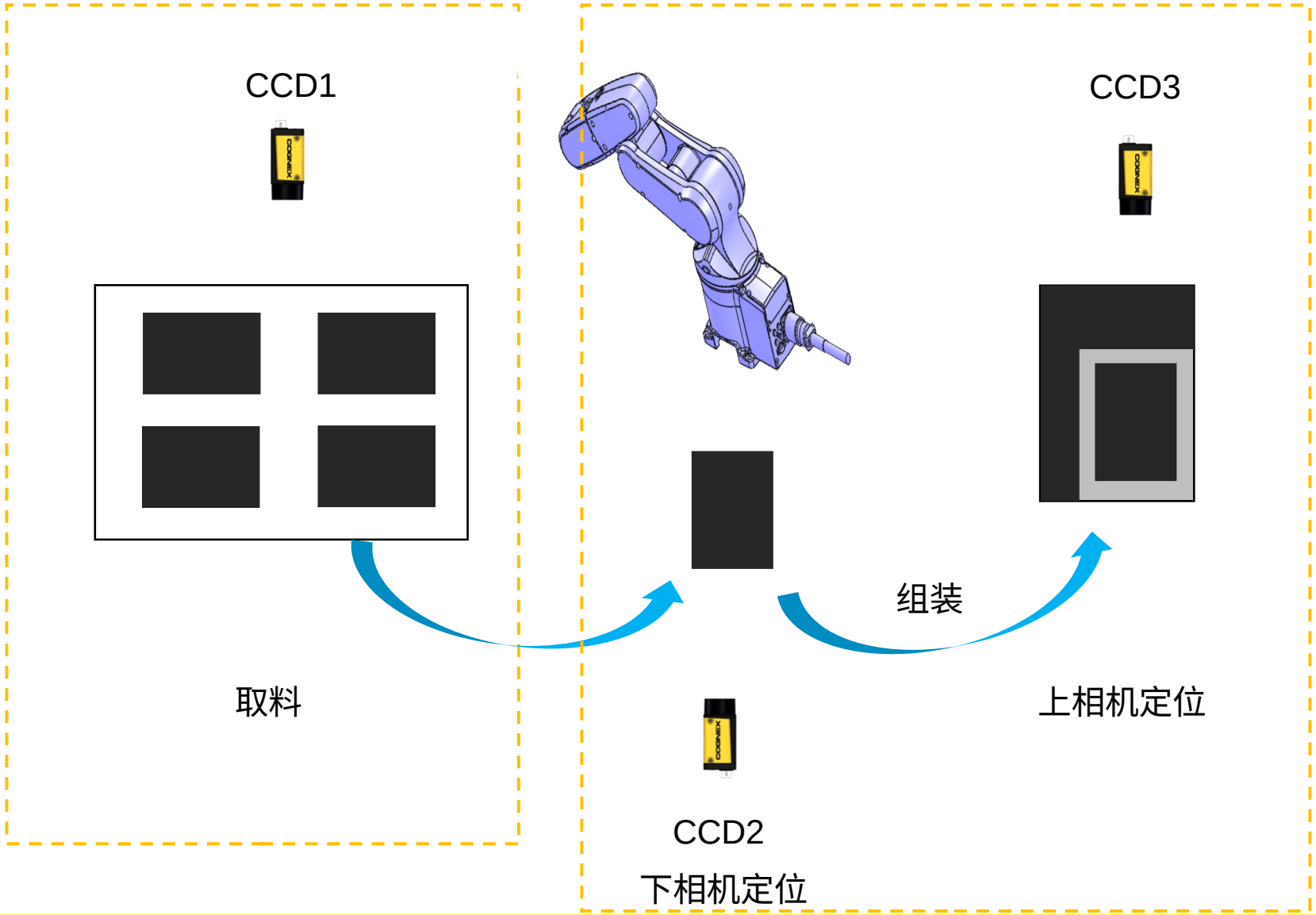
Machine 工作流程

CCD1 为大视野，需要单独标定



复检

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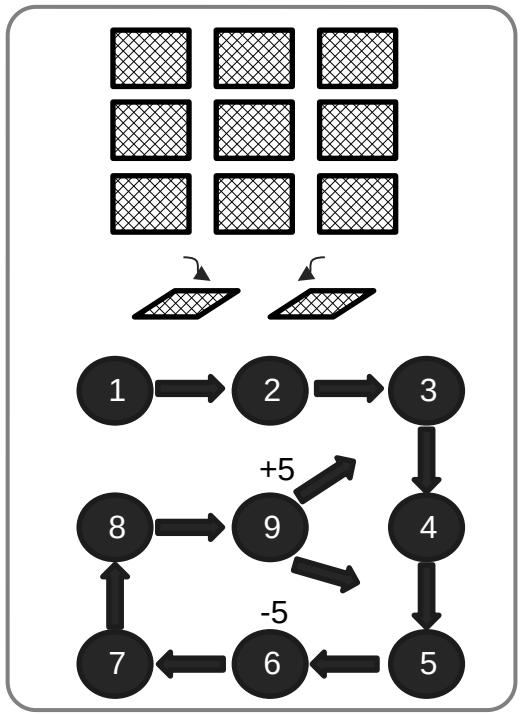
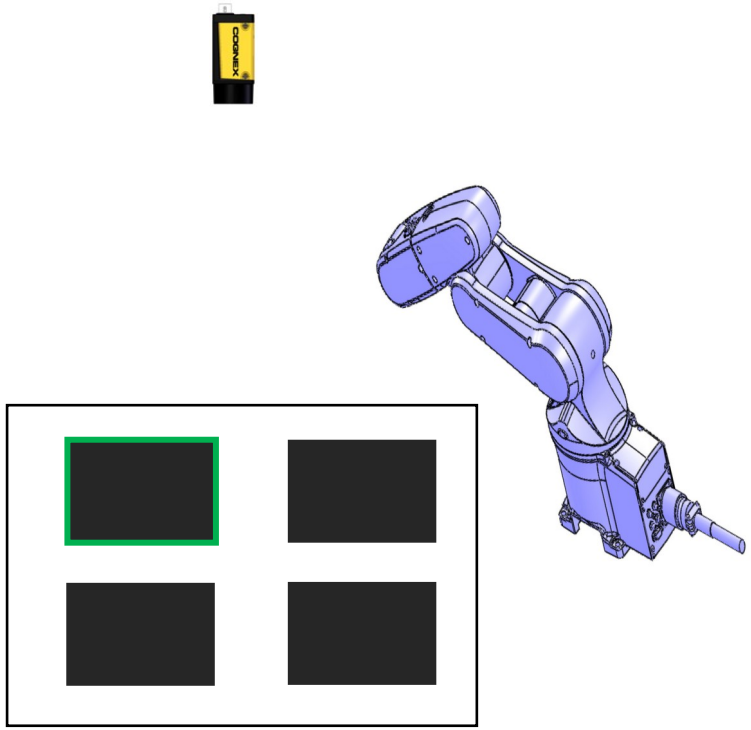
标定设计

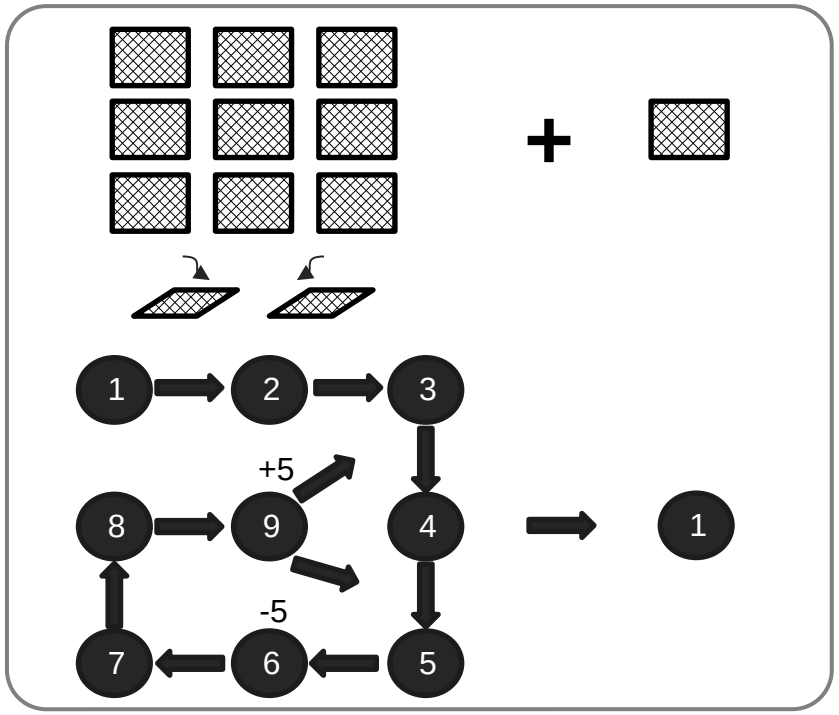
- 取料标定☒ CCD1 为大视野使用 4mm 标定片，需要确定 XYAngle 并且需要绑定 Robot 与产品的关系☒ 11point 标定 +Training 。
- 组装标定☒ CCD2 与 CCD3 使用 2mm 标定片，需要确定 XYAngel 并需要关联 CCD2 与 CCD3→ 联合标定 11+1point 标定。



11 Point Calibration



Training



11+1 Point Calibration

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标定通讯协议 – 取料标定

Machine Send	COGNEX Vision Send	OK/NG	Comment
SC,1,11	SC,1	OK	Start the calibration. SC: Start Calibration， 1: use first calibration file, 11： 执行 11 点标定
	SC,0	NG	
C1, X1,Y1,θ1	C1,1	OK	Send the P1 coordinate to the vision
	C1,0	NG	
C1, X2, Y2, θ2	C1,1	OK	Send the P2 coordinate to the vision
	C1,0	NG	
…	…	…	….
C1,X11,Y11,θ11	C1,1	OK	Send the P11 coordinate to the vision
	C1,0	NG	
EC	EC,1	OK	End the calibration
	EC,0	NG	
Train, X, Y, θ	Train, 1	OK	训练 Robot 取料时 Robot 的坐标
	Train, 0	NG	

- Use TCP/IP protocol, and the vision system is server, end by \r\n for each send and receive.

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标定通讯协议 – 对位标定

Machine Send	COGNEX Vision Send	OK/NG	Comment
SC,2,11,3,1	SC,1	OK	Start the calibration.
	SC,0	NG	
C2, X1,Y1,θ1	C1,1	OK	Send the P1 coordinate to the vision
	C1,0	NG	
C2 X2, Y2, θ2	C1,1	OK	Send the P2 coordinate to the vision
	C1,0	NG	
…	…	…	….
C2,X11,Y11,θ11	C1,1	OK	Send the P11 coordinate to the vision
	C1,0	NG	
C3, X, Y, θ	C3,1	OK	Send the P12 coordinate to the vision
	C3,0	NG	
EC	EC,1	OK	End the calibration
	EC,0	NG	

- Use TCP/IP protocol, and the vision system is server, end by \r\n for each send and receive.

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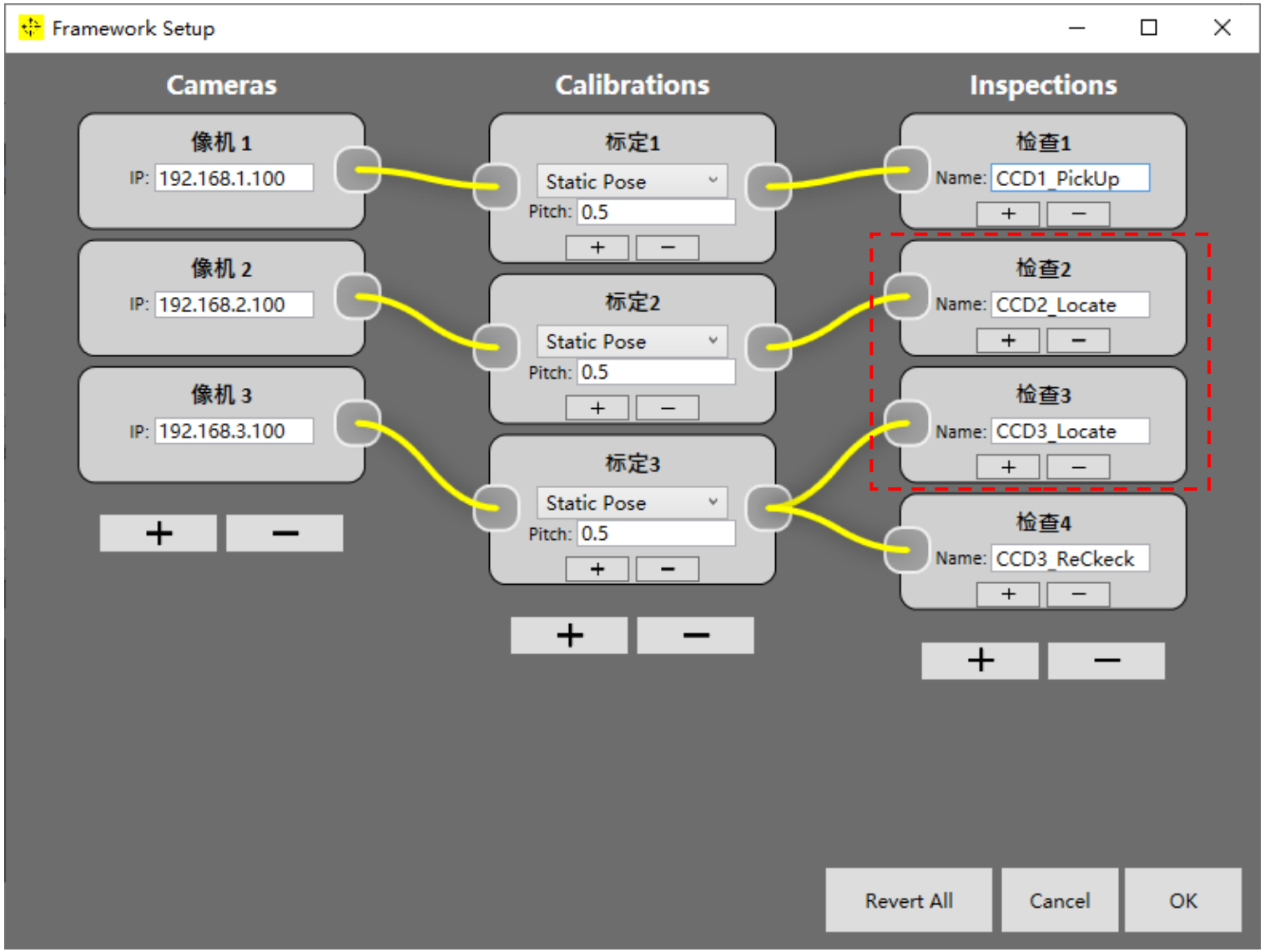
Framework 设计



取料

组装

复检

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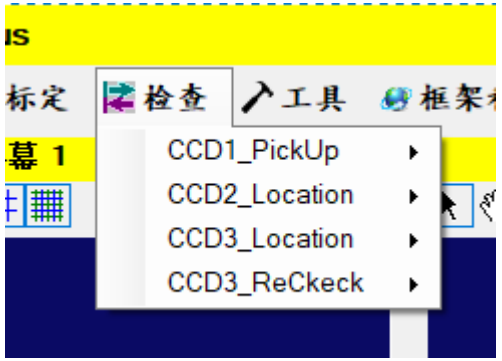
Inspection 设计

- CCD1_PickUp→ 训练 Robot 与生产时引导 Robot 取料
- CCD2_Locate→ 下相机定位
- CCD3_Locate→ 上相机定位
- CCD3_ReCheck→ 上相机复检



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生产通讯协议

Machine Send	COGNEX Vision Send	Comment
P0,SN,X,Y,Angle	P0,SN,Errorcode,X,Y,Angle	PickUp, SN: 当前产品的，Errorcode: 定义的错误代码，XYAgnle: 当前 Robot 的坐标。
P1, SN, X, Y, Angle	P1,SN,Errorcode,X,Y,Angle	产品 1 定位，SN: 当前产品的，Errorcode: 定义的错误代码，XYAgnle: 当前 Robot 的坐标。
P100, SN,X, Y, Angle	P100,SN,Errorcode,X,Y,Angle	产品 2 定位，SN: 当前产品的，Errorcode: 定义的错误代码，XYAgnle: 当前 Robot 的坐标。
P400,SN, X, Y2, Angle	P100,SN,Errorcode,Gap1,Gap2	产品 2 定位，SN: 当前产品的，Errorcode: 定义的错误代码，Gap: 测量的间隙。

- Use TCP/IP protocol, and the vision system is server, end by \r\n for each send and receive.

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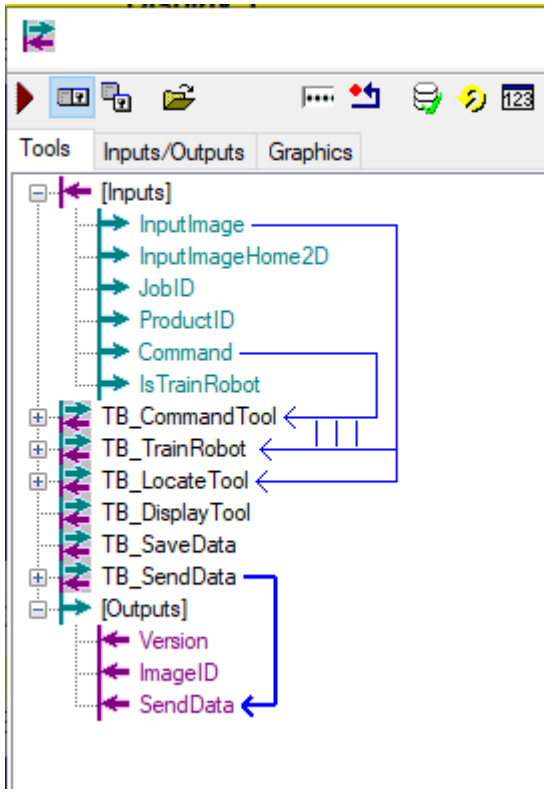
Inspection | CCD1_PickUp

- TB_CommandTool→ 解析指令
- TB_TrainRobot→ 训练 Robot 坐标
- TB_LocateTool→ 抓取产品特征
- TB_DisplayTool→ 将数据显示到 Display
- TB_SaveData→ 保存数据
- TB_SendData→ 将需要发送的数据整理到一起



Note： 先训练 Robot 取料的位置，再使用下列公式计算 Robot 的取料移动到的位置

DestinationRobotPose=DestinationPartPose.Compose(CurrentPartPose.Invert().Compose(CurrentRobotPose))

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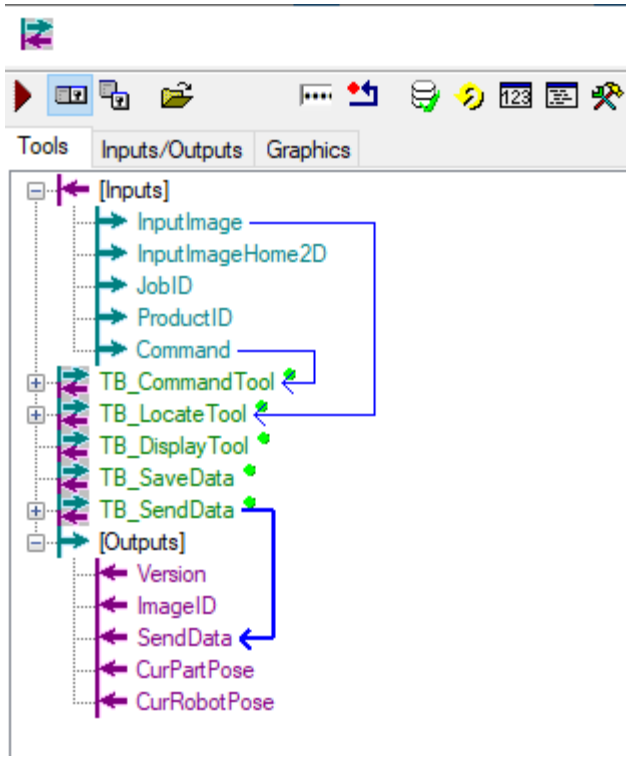
Inspection | CCD2_Location

- TB_CommandTool→ 解析指令
- TB_LocateTool→ 抓取产品特征
- TB_DisplayTool→ 将数据显示到 Display
- TB_SaveData→ 保存数据
- TB_SendData→ 将需要发送的数据整理到一起



Note： 在这个 Inspection 里面需要把当前的获取的产品位置与 Robot 的位置给到 CCD3_Location 的 Inspection 里面，用它们去计算 Robot 需要走到的位置。

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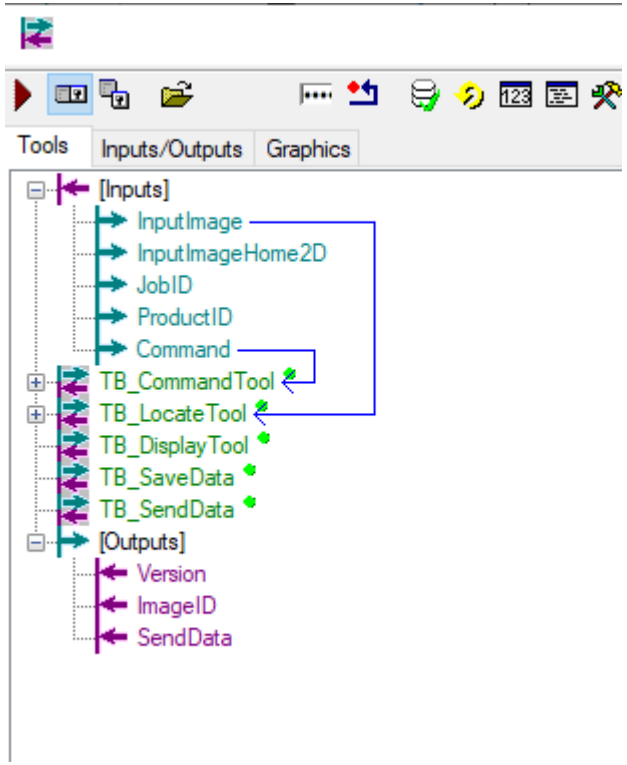
Inspection | CCD3_Location

- TB_CommandTool→ 解析指令
- TB_LocateTool→ 抓取产品特征并计算 Robot 需要移动到的位置
- TB_DisplayTool→ 将数据显示到 Display
- TB_SaveData→ 保存数据
- TB_SendData→ 将需要发送的数据整理到一起



Note： 使用下列公式计算 Robot 的需要移动到的位置

DestinationRobotPose=DestinationPartPose.Compose(CurrentPartPose.Invert().Compose(CurrentRobotPose))

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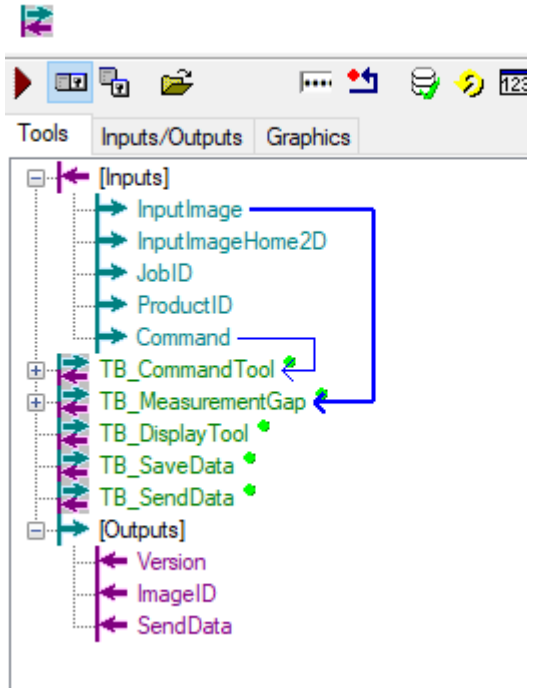
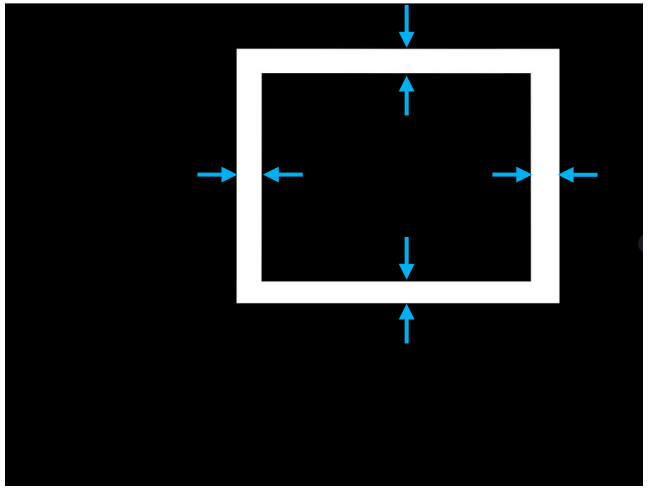
Inspection | CCD3_ReCheck

- TB_CommandTool→ 解析指令
- TB_MeasurementGap→ 测量贴合后产品的间的 Gap，每边都需要测量 10 组 Gap
- TB_DisplayTool→ 将数据显示到 Display
- TB_SaveData→ 保存数据
- TB_SendData→ 将需要发送的数据整理到一起





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Thank You

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